BTS –II(R)&I(S) –05.16 – 0779 Reg. No.

A

B. Tech. Degree II Semester Regular and I Semester Supplementary Examination May 2016

IT/CS/EC/CE/EE/ME/SE 1105 A / 1205 B BASIC MECHANICAL ENGINEERING
(*2015 Scheme*)

Time: 3 hours. Maximum Marks: 60

PART A
(Answer *ALL* questions)

($10 \times 2 = 20$)

I. (a) Explain equilibrium state of a system.
(b) Mention any four boiler mountings.
(c) Write short note on Otto cycle.
(d) Define capacity of refrigeration.
(e) What is the working principle of centrifugal pump?
(f) What do you mean by motion study?
(g) Differentiate annealing and normalizing?
(h) Explain any two welding defects.
(i) What are the differences between belt and chain drives?
(j) State two merits and two demerits of a nuclear power plant.

PART B

($4 \times 10 = 40$)

II. (a) Explain enthalpy. Derive the expression for change in enthalpy. (4)
(b) With a neat sketch explain Carnot cycle. A heat engine operates on Carnot cycle is supplied heat at the rate of 1700/ kJ/min and gives an output of 9 kW. Determine the thermal efficiency and the rate of heat rejection. (6)

OR

III. (a) Differentiate saturated steam and dry saturated steam. (4)
(b) What do you mean by compounding of steam turbines? With a neat sketch explain any two types. (6)

IV. (a) Differentiate two-stroke and four-stroke engine. (4)
(b) Explain any two types of lubricating system with a neat sketch. (6)

OR

V. (a) Derive expression for efficiency of diesel cycle. A diesel engine has a compression ratio of 15 and heat addition at constant pressure takes place at 6% of stroke. Find the air standard efficiency of the engine. Take γ for air as 1.4. (4)
(b) Explain summer and winter air conditioning systems, with neat sketches. (6)

(P.T.O.)

VI. (a) What are the differences between impulse turbine and reaction turbine? (5)

(b) How are pumps classified? With the help of neat sketch explain the working of reciprocating pump. (5)

OR

VII. (a) Explain the working of nuclear power plant, with a neat diagram. (5)

(b) Enumerate the contributions of F.W Taylor and Gilberth towards industrial engineering. (5)

VIII. (a) Explain brazing and soldering. (4)

(b) What are the different casting defects? Explain. (6)

OR

IX. (a) Explain with neat sketches. (4)

(i) Three high rolling mill (ii) Forward extrusion.

(b) Two pulleys, one 450 m diameter and the other 200 mm diameter, on parallel shafts 1,95 m apart are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. (6)

What power can be transmitted by the belt when the larger pulley rotates at 200 rpm, if the maximum permissible tension in the belt is 1 kN and the coefficient of friction between the belt and pulley is 0.25?

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